Portfolio and Investment Analysis: CAPM Regression Study

A comparative analysis of systematic risk and market sensitivity between Nestlé and Mitsubishi using Capital Asset Pricing Model regression with weekly excess returns.

Research Objective

To quantitatively assess systematic risk exposure and market sensitivity of two multinational corporations using CAPM regression framework with weekly excess returns.

Methodological Approach

Weekly adjusted closing prices converted to excess returns. Annual risk-free rate transformed to weekly basis. Ordinary least squares regression applied to CAPM model specification.

Mitsubishi Corporation

Beta coefficient: 1.386 ($p < 0.001$)

Coefficient of determination: $R^2 = 0.404$

Interpretation: Statistically significant market exposure with beta exceeding unity, indicating 38.6% greater volatility than market index. Explained variance of 40.4% suggests substantial systematic risk component.

Nestlé S.A.

Beta coefficient: $\beta \approx 0$ (statistically insignificant)

Coefficient of determination: $R^2 \approx 0$

Interpretation: Negligible market sensitivity with beta coefficient not statistically distinguishable from zero. Minimal explained variance indicates predominantly idiosyncratic risk structure and defensive portfolio characteristics.

Key Investment Implications

Mitsubishi demonstrates pronounced systematic risk and strong dependence on broad market movements, making it suitable for investors seeking market exposure. Conversely, Nestlé exhibits limited market sensitivity and lower systematic risk, positioning it as a defensive holding that may provide portfolio diversification benefits during market downturns.

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